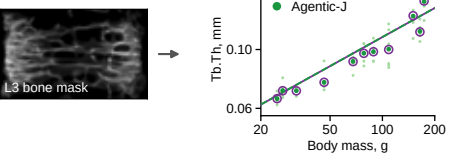
a **BoneJ** trabecular bone morphometry



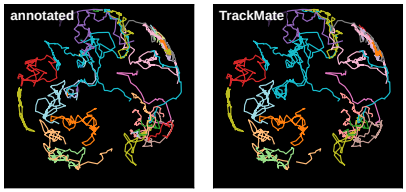
Tb.Th scales with body mass as published, slope 0.31 vs 0.31;
66 vertebrae: 3 of 4 metrics within 0.9%, Tb.Sp +5 to +10%

b **OrientationJ** orientation analysis

Placeholder

OrientationJ task: input image, result, one-line takeaway

c **TrackMate** mouse stem cell tracking



29 tracks over 101 frames, 1998 of 1998 spots on annotated cells;
mean speed 2.00 vs 2.02 µm/min annotated, r = 0.99

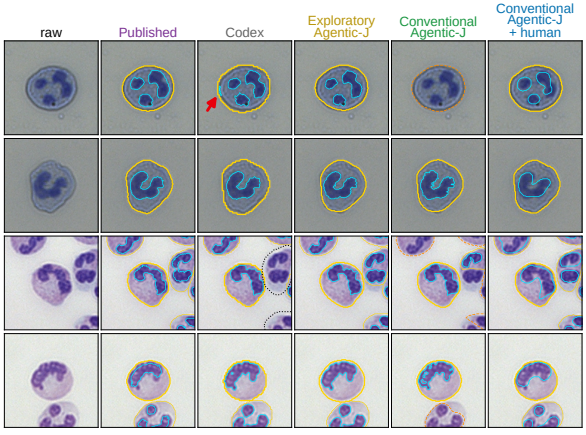
d

Analysis	Time	Scientific verdict	Biologist feedback
CD177 neutrophil morphology			
Published	weeks	no meaningful difference	reference segmentation
Codex CLI	13 min	0/30 features significant	many cells missed; faulty cell and nucleus segmentation
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	faulty nucleus segmentation; false-positive cells
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	cells segmented but discarded; nuclei missed or partly segmented
Conventional Agentic-J + human	488 min	0/11 tested significant 34 features extracted	clean cell segmentation; nuclei rounded and over-segmented

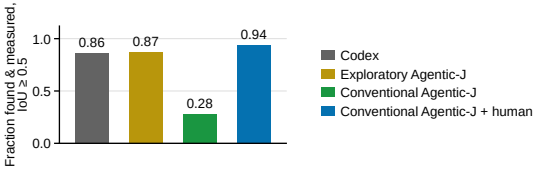
Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame	Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
Leishmania infection: TUNEL and proliferation		
not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL−, p = 0.0012
18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts p = 0.017; 10 of 15 parasites
22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
34 / 26 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	higher proliferation in TUNEL+ hosts p = 0.0012; all 15 parasites, 14 within 0.1

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

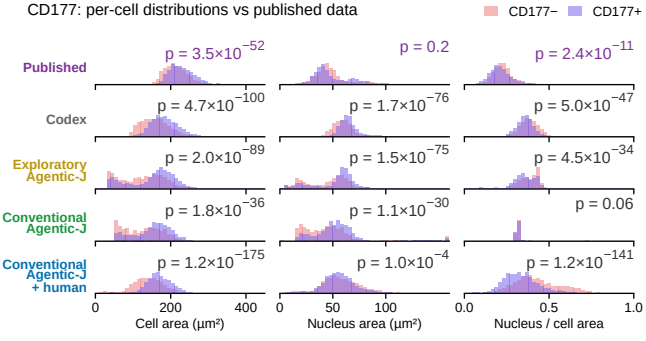
e CD177: segmentations of four example cells



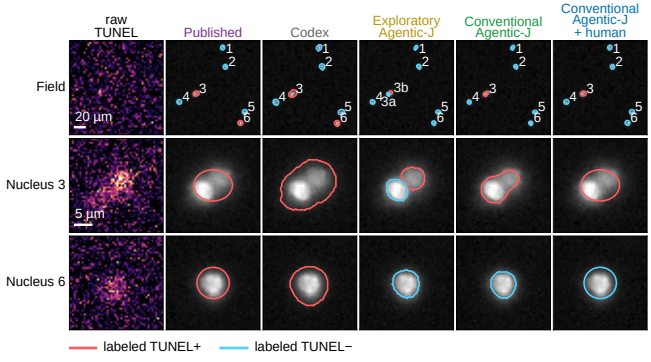
f CD177: published cells found and measured, n = 4389



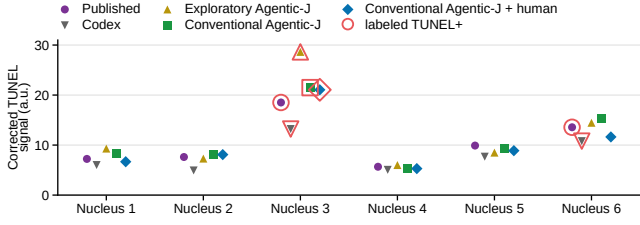
g CD177: per-cell distributions vs published data



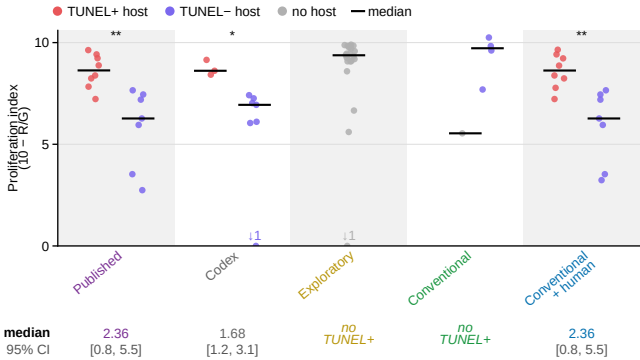
h Leishmania task 1: six nuclei, called by each pipeline



i Leishmania task 1: corrected TUNEL signal per nucleus



j Leishmania task 2: proliferation index by host TUNEL status



	Published	Codex	Exploratory	Conventional	Conventional + human
Δ median	2.36	1.68	*no TUNEL+*	*no TUNEL+*	2.36
95% CI	[0.8, 5.5]	[1.2, 3.1]			[0.8, 5.5]